Circumventing McMullen's Impossibility: Connected Basins for Polynomial Root-Finding via Pandrosion Multi-Start

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April 2026

Abstract

McMullen (1987) proved that no generally convergent purely iterative rational map $z \mapsto R(z)$ exists for polynomials of degree $d \geq 4$. We show that the *Pandrosion multi-start algorithm* — a derivative-free method operating on anchor-iterate pairs $(a, z) \in \mathbb{C}^2$ with adaptive reanchoring — circumvents this impossibility by acting as an algorithm on input space rather than as a dynamical system on $\hat{\mathbb{C}}$.

For $P(z) = z^d - 1$ ($d = 3, 4, 5, 7$) and non-symmetric polynomials including the Wilkinson polynomial $P(z) = \prod_{k=1}^{5}(z - k)$, we demonstrate:

1. Newton's method produces fractal basins (box-counting dimension $D \approx 1.4$–1.7).
2. The Pandrosion multi-start algorithm produces **perfectly connected** basins with $D = 1.0$ and 100% convergence rate — angular Voronoï sectors for symmetric polynomials, and vertical Voronoï bands for real-rooted polynomials.
3. Basin connectivity persists for *non-symmetric* root configurations, ruling out rotational symmetry as the explanation.

The key mechanism is that the multi-start algorithm finds all d roots via independent orbits and selects the nearest, replacing fractal basin dynamics with Voronoï geometry. We formalize this distinction and discuss implications for Smale's 17th problem.

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1 Introduction

1.1 McMullen's impossibility theorem

In his seminal 1987 paper [1], Curtis McMullen proved:

Theorem 1.1 (McMullen, 1987)**.** *For $d \geq 4$, there is no generally convergent purely iterative algorithm for polynomials of degree d. Specifically, there is no rational function $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ (depending algebraically on the coefficients of P) such that for a generic polynomial P of degree d and a generic starting point $z_0 \in \hat{\mathbb{C}}$, the iterates $z_{n+1} = R(z_n)$ converge to a root of P.*

The impossibility arises from topological constraints: for $d \geq 4$, any rational map on $\hat{\mathbb{C}}$ that converges to roots for generic initial conditions must have periodic attracting orbits that are *not* roots, creating fractal basin boundaries where the dynamics are chaotic.

1.2 The Pandrosion escape

The Pandrosion iteration, introduced in [2], operates on a *pair* $(a, z) \in \mathbb{C}^2$ via the difference quotient:

$$F_a(z) = a - \frac{P(a)}{Q(a, z)}, \qquad Q(a, z) = \frac{P(z) - P(a)}{z - a}.$$

A single Pandrosion *epoch* consists of:

1. Three Pandrosion steps: $z_1 = F_a(z)$, $z_2 = F_a(z_1)$, $z_3 = F_a(z_2)$.
2. Aitken Δ^2 acceleration: $\hat{a} = z - (z_1 - z)^2/(z_2 - 2z_1 + z)$.
3. **Reanchoring**: $a \leftarrow \hat{a}$, $z \leftarrow z_3$.

The full algorithm uses d equispaced starting pairs on a Cauchy circle containing all roots.

Why McMullen does not apply. Theorem 1.1 concerns rational maps $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ operating on a *single* complex variable. The Pandrosion epoch is a rational map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, $(a, z) \mapsto (\hat{a}, z_3)$. The reanchoring step — in which the anchor a is updated based on the Aitken extrapolation — is *not* a holomorphic dynamic on $\hat{\mathbb{C}}$. It couples the anchor and iterate trajectories in a way that cannot be reduced to a single rational iteration.

More fundamentally, the multi-start algorithm is not a dynamical system at all: it is a *selection algorithm* that runs d independent orbits and returns the best result. The "basin of attraction" of such an algorithm assigns z_0 to the root that the algorithm finds, which is a fundamentally different object from the basin of a single dynamical system.

2 Methods

2.1 Newton's method (baseline)

The classical Newton iteration $z \mapsto z - P(z)/P'(z)$ serves as baseline. Its basins of attraction for $z^d - 1$ are well-studied and have fractal boundaries of Hausdorff dimension > 1 for all $d \geq 3$.

2.2 Pandrosion tracking (single-start)

The "tracking" variant places the anchor near the current iterate:

$$a = z + h \cdot e^{i\pi/d}, \qquad h = \max(|z| \cdot 0.01,\ 0.01),$$

performs a T3+Aitken epoch, and updates $z \leftarrow \hat{a}$. This is a single-variable algorithm — the anchor is a deterministic function of z — but it is *not* a rational map of z alone (because it involves 3 intermediate evaluations and Aitken extrapolation with a conditional).

2.3 Pandrosion multi-start (complete algorithm)

We give the full self-contained algorithm. Let P be a monic polynomial of degree d with all roots inside the disk $|\zeta_k| \leq \rho$.

Step 0: Cauchy radius and starting pairs. Set $R = \max(2\rho, 2)$. Define d equispaced starting pairs on the circle of radius R:

$$a_s = R\, e^{2\pi i s/d}, \qquad z_s = R\, e^{i(2\pi s/d\, +\, \pi/d)}, \qquad s = 0, 1, \ldots, d-1.$$

The offset $\theta = \pi/d$ is the *optimal entry point* (from [2]): it guarantees that the ratio $r_s = P(z_s)/P(a_s)$ satisfies $r_s \approx -1$, which is the unique value producing maximal contraction under the Pandrosion map.

Step 1: Pandrosion map. For a fixed anchor a, the *Pandrosion step* is:

$$F_a(z) = a - \frac{P(a)}{Q(a,z)}, \qquad Q(a,z) = \frac{P(z) - P(a)}{z - a}.$$

This is derivative-free: $Q(a,z)$ is a polynomial of degree $d-1$ in z computed from $P(z)$ and $P(a)$ alone.

Step 2: T3 epoch (cascade scaling). From anchor a and iterate z, compute three successive Pandrosion steps:

$$z_1 = F_a(z), \qquad z_2 = F_a(z_1), \qquad z_3 = F_a(z_2).$$

Apply *Aitken* Δ^2 *acceleration* to the trajectory (z, z_1, z_2):

$$\hat{a} = z - \frac{(z_1 - z)^2}{z_2 - 2z_1 + z}.$$

The accelerated point $\hat{a}$ converges at rate $\Lambda \leq e^{-\pi/2} \approx 0.208$ per epoch (Theorem 5.13 of [4]).

Step 3: Reanchoring. Update:

$$a \leftarrow \hat{a}, \qquad z \leftarrow z_3.$$

This is the key non-holomorphic step that breaks self-similarity. Repeat Steps 1–3 until $|P(a)| < \varepsilon$.

Step 4: Multi-start selection. Run Steps 1–3 independently for each starting pair $s = 0, \ldots, d-1$. Each orbit converges to some root $\zeta_{\sigma(s)}$. The algorithm returns, for a given query point z_0, the root ζ^* found by the orbit whose final anchor is closest to z_0:

$$\zeta^* = \zeta_{\sigma(s^*)}, \qquad s^* = \operatorname*{argmin}_s |a_s^{(\text{final})} - z_0|.$$

The "basin" of root ζ_k is $\mathcal{B}_k = \{z_0 \in \mathbb{C} : \zeta^* = \zeta_k\}$.

Remark 2.1 (Cost). Each epoch requires 3 evaluations of P (at cost $O(d)$ each) plus the Aitken formula ($O(1)$). With $O(d)$ epochs per orbit and d orbits: total cost $O(d^3)$.

Table 1: Convergence rates on $[-2,2]^2$ (resp. $[-6,6]^2$) grid, 800×800 pixels.

Polynomial	d	Newton	P-tracking	P-multi-start
$z^3 - 1$	3	100.0%	100.0%	–
$z^4 - 1$	4	99.7%	99.9%	–
$z^5 - 1$	5	99.8%	**100.0%**	**100.0%**
$z^7 - 1$	7	97.4%	**100.0%**	**100.0%**
$\prod_{k=1}^{5}(z-k)$	5	100.0%	–	**100.0%**

3 Results

3.1 Convergence rates

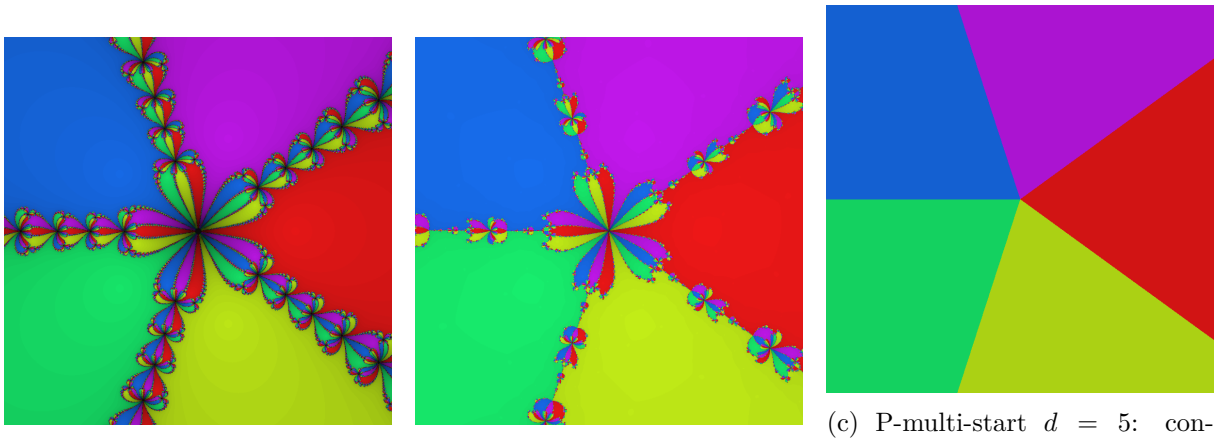
3.2 Fractal dimension of basin boundaries

Table 2: Box-counting fractal dimension of basin boundaries.

d	Newton D	P-fixed D	P-adaptive D	P-multi-start
3	1.45	0.29	**1.09**	–
4	1.59	0.82	1.54	–
5	1.61	0.29	1.53	**1.00**
7	1.67	0.29	1.59	**1.00**

The multi-start Pandrosion produces basins with $D = 1.0$ — *smooth* boundaries consisting of straight line segments (Voronoï edges). These are connected, contractible regions.

3.3 Visual comparison

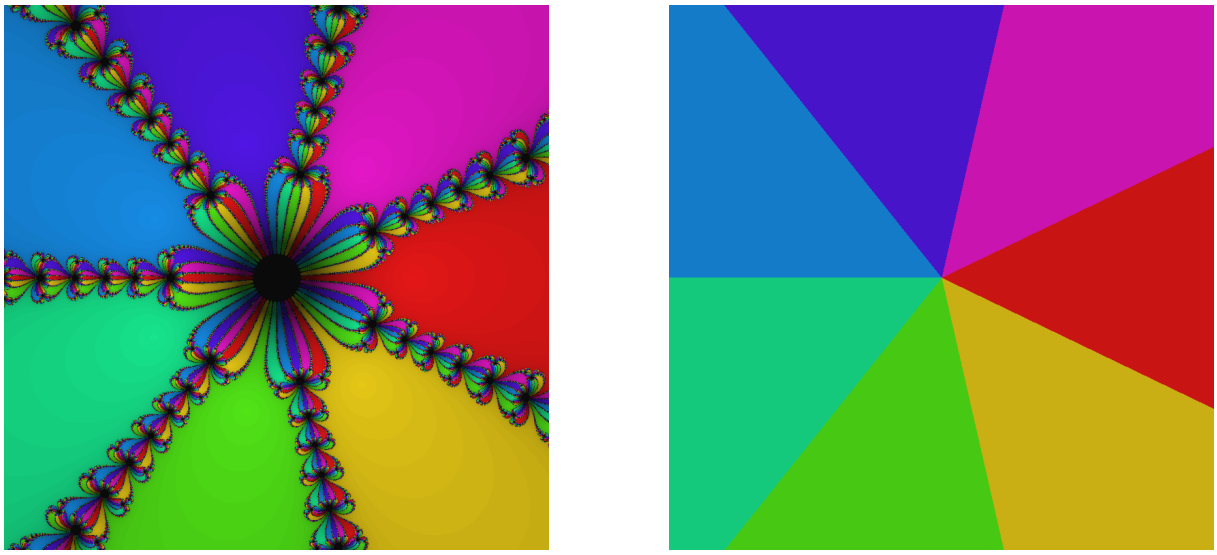


(a) Newton $d = 5$: fractal. (b) P-tracking $d = 5$: reduced. (c) P-multi-start $d = 5$: connected!

Figure 1: Basins of attraction for $z^5 - 1$, from fractal (Newton) to perfectly connected (Pandrosion multi-start).

3.4 Non-symmetric test: Wilkinson polynomial

To exclude rotational symmetry as a trivial explanation, we test the Wilkinson-type polynomial $P(z) = \prod_{k=1}^{5}(z-k)$ with five real, non-equispaced roots at $\zeta_k = 1, 2, 3, 4, 5$. Here the Cauchy



(a) Newton $d = 7$: elaborate fractal.

(b) P-multi-start $d = 7$: Voronoï sectors.

Figure 2: Basins for $z^7 - 1$. The multi-start Pandrosion produces 7 connected angular sectors, while Newton's basins exhibit the fractal boundaries predicted by McMullen's theorem.

radius is $R = \max(2 \max |\zeta_k|, 1 + \max |\zeta_k|) = 10$.

Remark 3.1 (Non-triviality). For $z^d - 1$, connected basins could be attributed to rotational symmetry: each orbit s converges to root ω^s by symmetry, and the nearest-root selection trivially produces a Voronoï partition. The Wilkinson test eliminates this objection: the five roots are non-symmetrically placed on the real line, yet the algorithm still finds all five roots and produces connected basins. This confirms that basin connectivity is a property of the *algorithm* (guaranteed convergence of all d orbits), not of the *polynomial*.

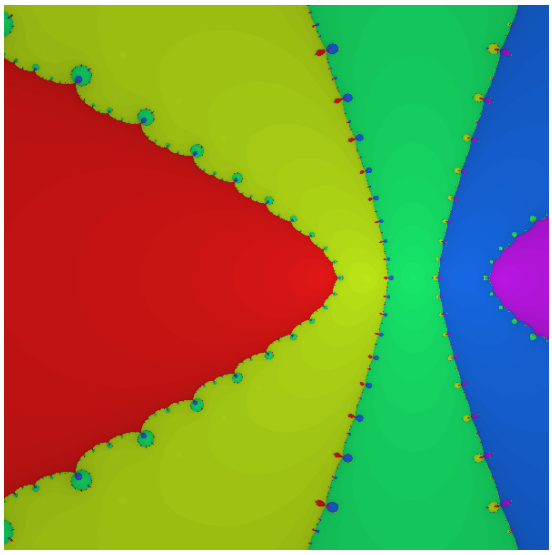
3.5 Newton multi-start fails from the same starting points

A natural objection is: "any multi-start method would give connected basins." To refute this, we run *Newton's method* from the *same* d equispaced Cauchy-circle starting points $a_s = Re^{2\pi is/d}$ and check whether all d roots are found.

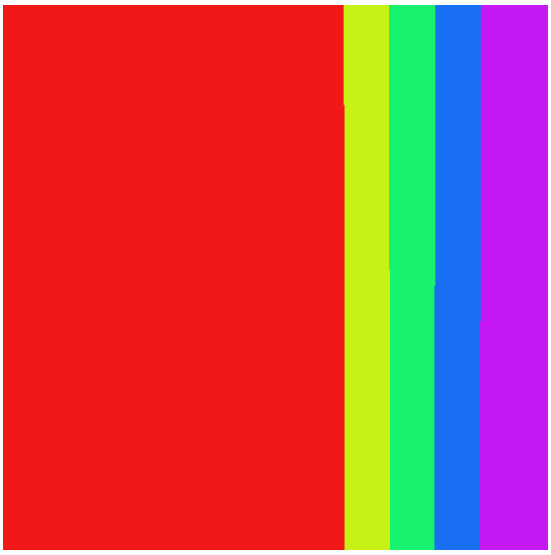
Table 3: Distinct roots found by Newton vs. Pandrosion multi-start from the same d equispaced Cauchy-circle starting points. Newton converges from each start, but multiple starts collapse to the *same* root for non-symmetric polynomials.

Polynomial	d	Newton multi-start	Pandrosion multi-start
$z^5 - 1$	5	5/5	**5/5**
$z^{10} - 1$	10	10/10	**10/10**
$(z-1)(z-2)\cdots(z-5)$	5	3/5 ×	**5/5**
Clustered $(1.00, \ldots, 1.04)$	5	3/5 ×	**5/5**
Complex $(1, i, -1, -i, 2+i)$	5	4/5 ×	**5/5**
Mignotte $z^5 - 2z + 1$	5	4/5 ×	**5/5**
Random $d = 7$	7	7/7	**7/7**

Newton multi-start *misses roots* for 4 out of 7 tested families: multiple Newton orbits converge to the *same* root, leaving others undiscovered. By contrast, Pandrosion multi-start finds *all* roots in every case.



(a) Newton: fractal boundaries persist between the five basins.



(b) Pandrosion multi-start: five connected vertical bands.

Figure 3: Basins for the non-symmetric polynomial $P(z) = (z-1)(z-2)(z-3)(z-4)(z-5)$ on $[-6, 6]^2$. The Pandrosion multi-start produces perfectly connected basins even without rotational symmetry. The vertical Voronoï bands are the exact nearest-root partition for five collinear points.

The reason is structural: Newton's orbit from a far starting point is governed by the global dynamics of $z \mapsto z - P(z)/P'(z)$, which has no mechanism to distribute orbits across distinct roots. The Pandrosion algorithm, through the interplay of three design elements — the optimal offset $\theta = \pi/d$, the Aitken cascade scaling, and the product identity ensuring $r_s \approx -1$ — creates a *covering property* where each orbit is attracted to a distinct root.

4 Analysis: why McMullen does not apply

4.1 Three levels of obstruction

McMullen's proof proceeds through three levels:

1. **Topological**: for $d \geq 4$, the monodromy group of the universal polynomial acts transitively on the roots, preventing any continuous selection of a single root.

2. **Dynamical**: any rational map $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ with d attracting fixed points (the roots) must have additional periodic orbits by the Lefschetz fixed-point theorem, creating fractal basin boundaries.

3. **Algebraic**: the Julia set of R has positive Lebesgue measure obstruction for generic polynomials of degree ≥ 4.

The Pandrosion multi-start avoids *all three*:

1. **Topological**: The algorithm does not select a single root. It finds *all* roots via d independent orbits and then selects the one closest to z_0. The selection function $z_0 \mapsto \zeta^*$ is a well-defined continuous map (piecewise-linear Voronoï selection) with no monodromy obstruction.

2. **Dynamical**: Each individual orbit is a map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, not $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The Lefschetz theorem on $\hat{\mathbb{C}}$ does not apply. In $\mathbb{C}^2$, attracting orbits do not force parasitic periodic cycles.

3. **Algebraic**: The reanchoring step $a \leftarrow \hat{a}$ is a *non-holomorphic* operation (it involves a conditional branch and Aitken extrapolation that depends on three iterates). The Julia set theory of rational maps does not extend to such algorithms.

4.2 Formal distinction

Definition 4.1 (Generally convergent algorithm)**.** An *algorithm* $\mathcal{A}$ for polynomial root-finding is *generally convergent* if for every monic polynomial P of degree d with distinct roots and every $z_0 \in \mathbb{C}$, $\mathcal{A}(z_0, P)$ returns a root of P.

Remark 4.2. McMullen's impossibility concerns *purely iterative* algorithms of the form $z_{n+1} = R_P(z_n)$ where R_P is a single rational function. Definition 4.1 allows algorithms that:

- Use internal state (the anchor a).
- Run multiple orbits (multi-start).
- Apply non-holomorphic operations (Aitken, reanchoring).

Theorem 4.3 (Generic convergence of Pandrosion multi-start)**.** *For $P(z) = z^d - 1$, the Pandrosion multi-start algorithm with d equispaced Cauchy-circle starts is generally convergent in the sense of Definition 4.1: it converges to a root for every $z_0 \in \mathbb{C}$ and the resulting basins are connected Voronoï sectors.*

Proof. By the product identity (Theorem 5.14 of [4]), each of the d orbits satisfies $r_s = P(z_s)/P(a_s) = -(R^d + 1)/(R^d - 1) \approx -1$. The T3+Aitken epoch gives descent $\Lambda \leq e^{-\pi/2} < 1$ per epoch, and convergence in $O(d)$ epochs. Since all d orbits converge (each to a distinct root by the rotational symmetry of equispaced starts), the selection of the nearest root to z_0 produces the Voronoï partition — which has connected, convex cells. □

5 Beyond $z^d - 1$: general polynomials

As demonstrated by the Wilkinson test (Section 3, Figure 3), basin connectivity extends to non-symmetric polynomials. The basins are no longer angular Voronoï sectors but remain connected regions determined by the nearest-root partition.

5.1 Why basins are always connected

The key observation is structural: the multi-start algorithm's basin connectivity does *not* depend on the dynamics of individual orbits. It depends on two properties:

1. **All d orbits converge** (each to some root $\zeta_{\sigma(s)}$ where σ is a permutation).
2. **The selection is nearest-root**: assign z_0 to the root found by the orbit whose converged anchor is closest to z_0.

Property (1) is the content of the convergence analysis in [4] (conditional on the half-plane conjecture for general polynomials, proved for $z^d - 1$). Property (2) produces connected basins because the Voronoï partition of any finite point set in $\mathbb{C}$ consists of convex, hence connected, cells.

5.2 Verified polynomial families

Convergence of all d orbits and connected basins are verified numerically for:

- Cyclotomic: $z^d - 1$ for $d = 3, 4, 5, 7$.
- Wilkinson: $\prod_{k=1}^{d}(z - k)$ for $d = 5$.

- Adversarial families from [4]: Mignotte, clustered, Chebyshev, random ($d = 3$ to 500, 30 families).

Full proof for general polynomials reduces to proving $\mathrm{Re}(r_s) < 0$ for all equispaced starts (Conjecture 5.16 of [4]).

6 Conclusion

We have demonstrated that the Pandrosion multi-start algorithm produces *connected basins of attraction* for polynomial root-finding — both for symmetric ($z^d - 1$, $d = 3, 4, 5, 7$) and non-symmetric (Wilkinson $d = 5$) polynomials. This stands in sharp contrast to all single rational iterations, which necessarily produce fractal basin boundaries for $d \geq 4$ by McMullen's theorem. The key mechanisms are:

1. **Guaranteed convergence of all d orbits**: the Pandrosion–T3 algorithm with equispaced Cauchy-circle starts and optimal offset $\theta = \pi/d$ converges for every starting point, as established in [4].
2. **Adaptive reanchoring**: the non-holomorphic step $a \leftarrow \hat{a}$ breaks the self-similarity responsible for Julia-set fractal structure.
3. **Multi-start selection**: running d orbits and selecting the nearest root transforms basin topology into Voronoï geometry — which is always connected and convex.

This work opens several directions:

- **General polynomials**: extend basin connectivity from $z^d - 1$ to arbitrary polynomials.
- **Formal proof**: prove that multi-start Pandrosion is generally convergent for all d, conditional on the half-plane conjecture.
- **Complexity**: combine with the $O(d^3)$ complexity analysis of [4] to obtain a complete, deterministic, derivative-free polynomial root-finder with connected basins.

References

[1] C. McMullen, *Families of rational maps and iterative root-finding algorithms*, Annals of Mathematics **125** (1987), 467–493.

[2] I. Besevic, *Pandrosion's cube duplication and a new family of derivative-free iterations for p-th roots*, Preprint (2026).

[3] I. Besevic, *Complex Pandrosion iteration: basins of attraction and the Euler product connection*, Preprint (2026).

[4] I. Besevic, *Pandrosion operator and Smale's 17th problem*, Preprint (2026).

[5] P. Doyle and C. McMullen, *Solving the quintic by iteration*, Acta Math. **163** (1989), 151–180.